

scverse x cellpainting 2026

Uncertainty quantification for Cell Painting and Omics Data

Yunhee Jeong

Bayer Crop Science

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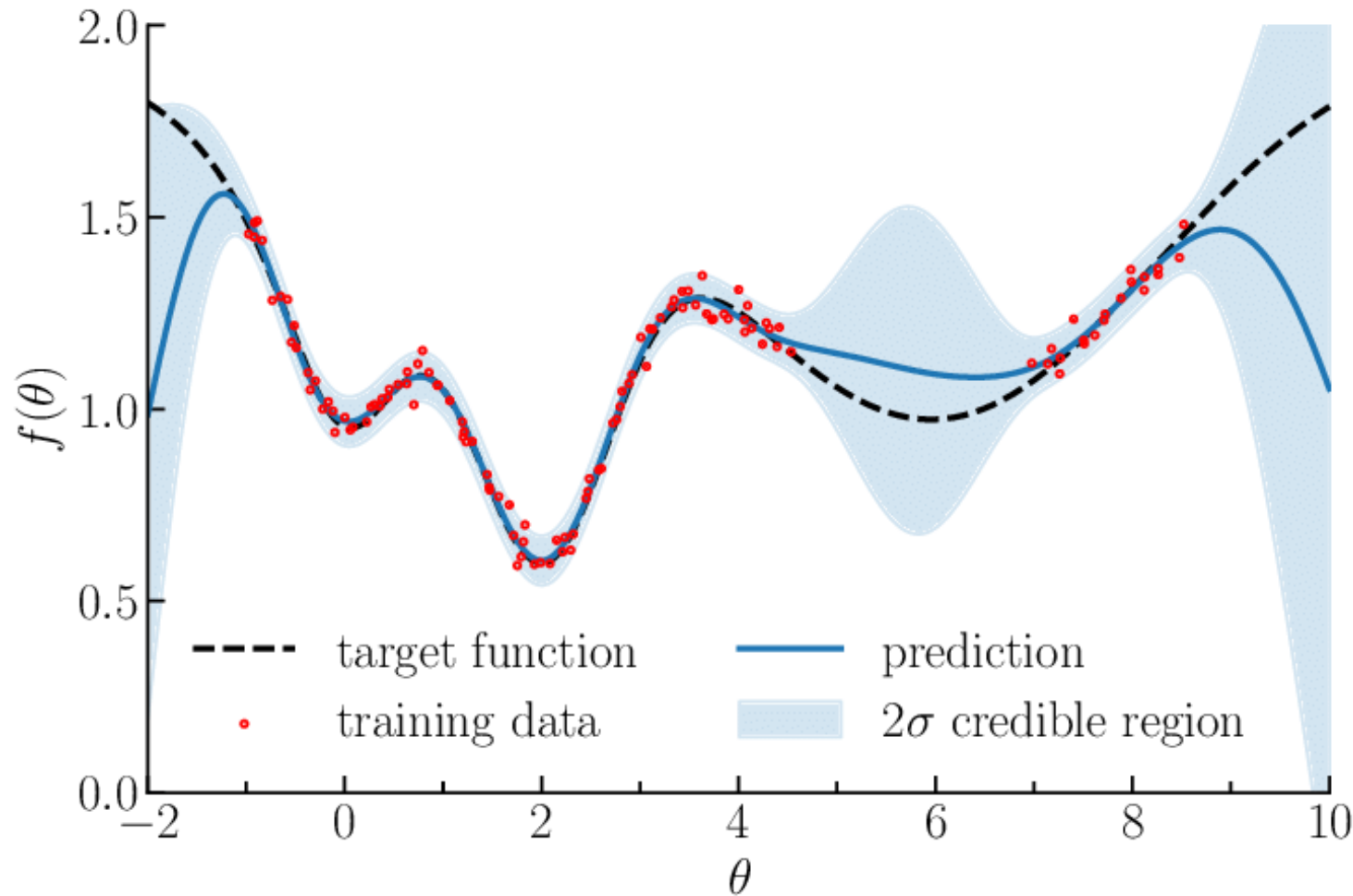
$$\hat{y} = f(x; \theta)$$

How much do we trust the prediction?

How (un)certain are we about the prediction?

- How often multiple models output the same prediction for the given input?
- How large the confidence interval is?
- Is the input data from the training data distribution?
- How noisy the input is?

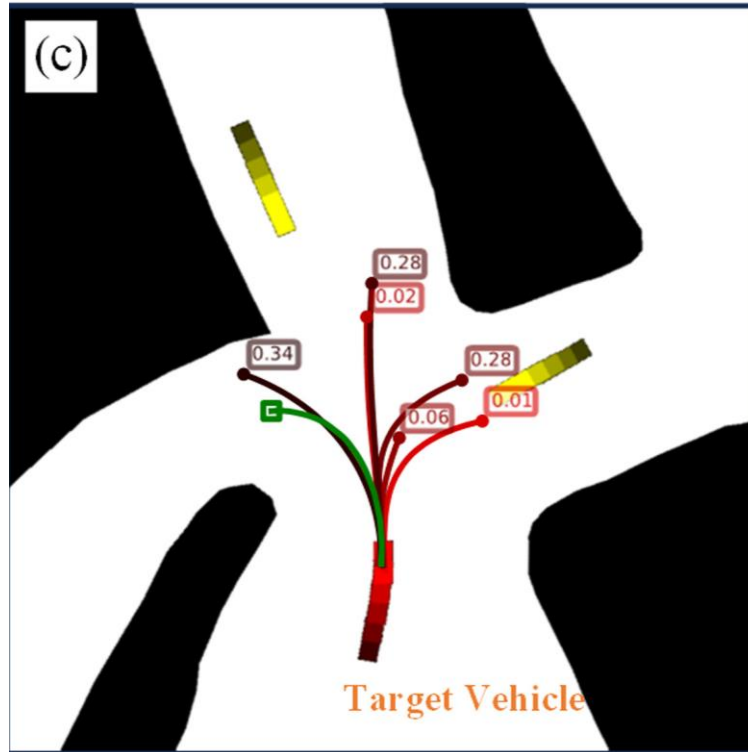
How do we handle the uncertainty?



Common approaches

- **Multiple predictions**
 - Ensemble
 - MC dropout
- **Distribution estimation**
 - Bayesian approach
 - Distribution parameter estimation
- **Interval estimation**
 - Quantile regression
 - Conformal prediction

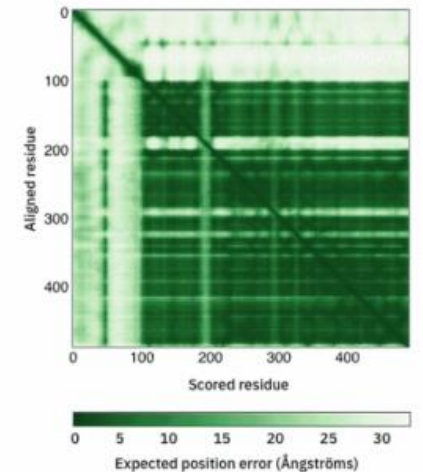
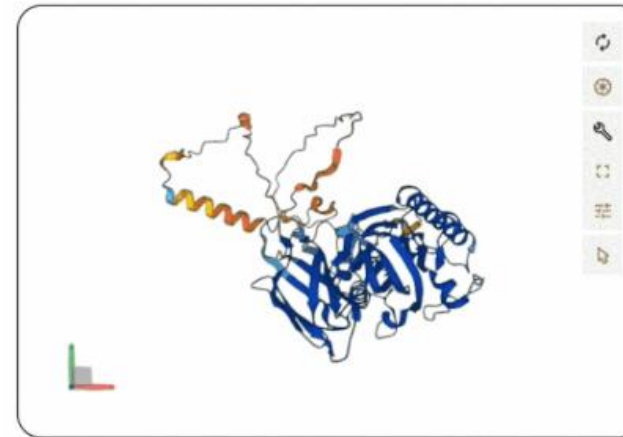
How do people use UQ modeling?



Trajectory determination in autonomous driving

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Sphingomyelinase



Model Confidence ⓘ

Very high (pLDDT > 90) Low (70 > pLDDT > 50)
High (90 > pLDDT > 70) Very low (pLDDT < 50)

Confidence measure in *AlphaFold*



Why is it important for biomedical research?

- Chemical space and biological variations are incredibly large
- Data generation cost is very high
- We will (very likely) have insufficient data to build a well-generalized model
- Our models' predictions might not always be fully credible!

Chemical space

Chemical space refers to the set of all molecules that could possibly exist.² The size of drug-like chemical space is subject to debate and has been estimated to lie anywhere between 10^{23} and 10^{180} compounds depending on the way it is calculated, as explained in Polishchuk et al.³ However, an oft-cited middle ground for the number of synthetically accessible small organic compounds is 10^{60} .⁴ This figure is based on a molecule that contains 30 atoms (C, N, O, or S), 4

What does a typical human genome sequence look like with respect to the variants that it contains? On average, compared to a reference human genome, a person's ~6 billion-nucleotide genome sequence will have:

- ~5,000,000 SNVs that involve ~5,000,000 nucleotides
- ~600,000 insertion/deletion variants that involve ~2,000,000 nucleotides
- ~25,000 structural variants that involve >20,000,000 nucleotides.

Goals of our workstream

- Understand probabilistic theory behind uncertainty quantification
- Apply uncertainty quantification methods to cell painting/omics data downstream analyses
 - Lightning-UQ-Box
 - AnnData
 - Scvi-tools
- Present your work at the end of the event!

References

- <https://lightning-uq-box.readthedocs.io/en/latest/>
- <https://anndata.readthedocs.io/en/latest/>
- <https://docs.scvi-tools.org/en/1.3.3/index.html>